

Shivam Gupta

Full-Stack Software Engineer

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PROFESSIONAL SUMMARY

Results-driven Full-Stack Software Engineer with 4+ years of experience designing, building, and deploying production-grade distributed systems. Expertise in React.js, TypeScript, Node.js, NestJS, Python, and PostgreSQL. Proven track record of architecting algorithmic trading platforms, autonomous AI agents, real-time communication SDKs (WebRTC/Mediasoup), and native desktop applications. Skilled in Docker, CI/CD, monorepo tooling (Nx/Turborepo), and microservices. Represented organization at ATR Open House 2025, Kyoto, Japan. Passionate about clean architecture, system design, and shipping impactful software.

TECHNICAL SKILLS

- **Languages:** JavaScript, TypeScript, Python, HTML5, CSS3, SQL
- **Frontend:** React.js, Next.js, Vite, Tailwind CSS, Chakra UI, D3.js, Recharts, React Query
- **Backend:** Node.js, NestJS, Express.js, Fastify, FastAPI, REST APIs, GraphQL, Socket.io, BullMQ
- **Databases:** PostgreSQL, MySQL, Redis, SQLite, Prisma ORM, TypeORM, Qdrant (Vector DB)
- **DevOps & Tools:** Docker, Docker Compose, Nx, Turborepo, Git, GitHub Actions, CI/CD, Nginx, Firebase
- **Real-Time & AI:** WebRTC, Mediasoup SFU, WebSockets, LLM Integration (Gemini, OpenRouter), Groq Whisper, TTS/STT Pipelines
- **Other:** Agile/Scrum, Monorepo Architecture, Microservices, PWA, OAuth/JWT/RBAC, AES-256 Encryption, OCR (PaddleOCR)

PROFESSIONAL EXPERIENCE

Full-Stack Software Engineer | Wisflux Tech Labs | Jaipur, India

Jun 2022 – Present

- Engineered multi-service SaaS applications using React.js, TypeScript, NestJS, and PostgreSQL, serving multiple production modules with measurable performance improvements.
- Architected modular backend services with JWT authentication, role-based access control (RBAC), and API versioning, reducing API response latency by up to 30%.
- Represented Wisflux at ATR Open House 2025 in Kyoto, Japan — demonstrated real-time offline communication solutions and engaged in international technical collaborations.
- Built FLOS, a disaster-response communication tool using Mediasoup SFU, WebSockets, and Automatic Speech Recognition (ASR) for hands-free emergency communication in low-connectivity environments.
- Dockerized development and production environments; streamlined CI/CD pipelines, accelerating developer onboarding by 40% and improving release stability.
- Developed and published Streamverse, an open-source NPM package enabling WebRTC-based audio, video, and screen sharing with Mediasoup SFU integration.

- Implemented real-time data synchronization layers using Socket.io and WebSockets across multiple client-facing applications.

KEY PROJECTS

Smart Trading Platform

Tech: React, Vite, NestJS, Prisma, Python FastAPI, PostgreSQL, Socket.io, Docker, Nx Monorepo

- Designed and built a full-stack algorithmic trading platform for the Indian stock market (NSE) supporting 28 backtested strategies across intraday and swing timeframes.
- Implemented live market scanning (60-second intervals), interactive candlestick charting (D3.js), automatic portfolio tracking, and real-time WebSocket + Telegram alert notifications.
- Architected as an Nx monorepo with a React + Vite frontend, NestJS API, Python FastAPI backtesting engine, and PostgreSQL database — all containerized with Docker Compose.

Echo — Encrypted PWA Messenger

Tech: React, TypeScript, NestJS, Socket.io, WebRTC, Prisma, PostgreSQL, Docker, Turborepo

- Built a production-grade, end-to-end encrypted messaging PWA with real-time chat, voice messages, image sharing, and online status tracking using Socket.io.
- Implemented peer-to-peer encrypted communication, service worker offline support, and cross-device synchronization in a Turborepo monorepo architecture.

AgentSystem — Autonomous AI Developer Agent

Tech: Next.js, Fastify, BullMQ, Redis, Qdrant, OpenRouter, Docker, Turborepo

- Developed an autonomous AI agent platform with a Next.js dashboard streaming real-time logs via Socket.io, powered by a Fastify API gateway and BullMQ task queues.
- Integrated Qdrant vector database for long-term context memory, enabling persistent agent knowledge across sessions.

Jarvis — Voice-First AI Desktop Assistant

Tech: Tauri, Rust, React, TypeScript, Groq Whisper, Streaming LLM, TTS Pipeline

- Engineered a native, voice-first AI desktop assistant for Linux using Tauri + Rust, achieving sub-second speech-to-text via Groq Whisper and sentence-chunked TTS playback.
- Implemented barge-in interruption handling and parallel background task execution without blocking the conversational flow.

Glacier Dev — AI Web Development Platform

Tech: Next.js 14, Google Gemini, OpenRouter, Supabase, NextAuth, Monaco Editor, Sandpack

- Created an autonomous AI-powered web development platform using multi-agent orchestration (Architect, Backend, UI agents) powered by Gemini 2.5 Flash.
- Integrated Monaco code editor, Sandpack live preview, and one-click Vercel deployment for seamless prompt-to-production workflows.

EDUCATION

Bachelor of Technology (B.Tech), Computer Science Engineering

Anand International College of Engineering, Jaipur | 2019 – 2023 | CGPA: 8.5 / 10.0

ACHIEVEMENTS & RECOGNITION

- Represented Wisflux Tech Labs at ATR Open House 2025, Kyoto, Japan — international technology showcase and collaboration event.
- Published open-source NPM package (Streamverse) for real-time WebRTC communication with Mediasoup SFU.
- Maintain 55+ repositories on GitHub spanning full-stack web, AI/ML, algorithmic trading, and native desktop applications.